

A Mechanistic Model of Droplet Formation in Matsuo’s Peptide-Coacervate System

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Abstract

Matsuo and Kurihara’s [1] peptide-coacervate system converts a disulfide precursor M_{pre} into monomer M , which oligomerizes into short peptide chains, which phase-separate into liquid droplets. This note gives a model that connects those two stages: a mechanistic reaction-network model of the oligomerization step, feeding a phase-separation model of droplet formation whose response climbs toward completion gradually rather than instantaneously. All fits below use the paper’s own Source Data files (see §1.1). Fit jointly to all three available real curves — two DTT concentrations, two figures — the oligomerization stage reproduces every one of them with a single shared set of rate constants ($R^2 = 0.990\text{--}0.996$); a separate, genuinely held-out test — fitting the 25 mM data alone under the naive assumption that DTT enters at first order, then predicting the 13 mM curve with zero new parameters — gives $R^2 = 0.61$, which the joint fit above resolves by identifying a DTT reaction order of $p \approx 2.37$ instead. The droplet-formation stage reproduces the full rise-peak-decline shape of the droplet-conversion curve essentially exactly ($R^2 = 0.9997$).

1 The system and the two observables

Reduction of M_{pre} by DTT releases monomer M , which oligomerizes into peptide chains P_n , which associate with BnSH and undergo liquid–liquid phase separation into droplets Q . Two time-courses are measured, and the two need different treatment before either can be fit.

Observable 1: residual M_{pre} , by a fluorescamine assay. This one needs a derivation before it can be used: fluorescamine detects free primary amines, not the disulfide bond DTT reduces (a different functional group), so it cannot see M_{pre} ’s own reduction step. What it tracks is peptide-bond formation: each new bond consumes exactly one free amine (the incoming monomer’s) while leaving the growing chain’s own N-terminal amine untouched, so the measured signal is exactly the total *particle count* — monomer plus unreacted oligomer chains — not residual M_{pre} itself. Writing $x = [M_{\text{pre}}]$, $m = [M]$, $p_n =$ concentration of oligomer P_n , the correct target for this curve is

$$O(t) = \frac{2x(t) + m(t) + \sum_{n=2}^N p_n(t)}{2x(0)} \quad (1)$$

(the factor of 2: each M_{pre} carries two amines) — not $x(t)$ itself.

Observable 2: conversion to droplet material, $q(t)$, by ^1H NMR. This one needs no such derivation — it is reported directly as the fraction of starting material converted to droplets, so $q(t)$ is already the quantity the model must predict, no translation required. Its model, (7)–(9) below, is the thing being fit to it; there is no separate formula defining what $q(t)$ is, the way (1) was needed for observable 1.

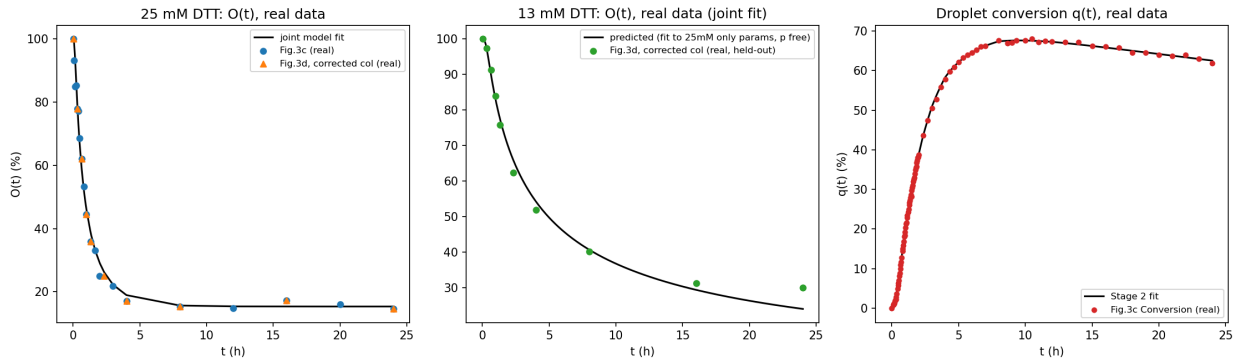
1.1 The source data

All data fit in this note come from Matsuo & Kurihara (2021) [1], *Nat. Commun.* 12:5487, DOI 10.1038/s41467-021-25530-6 (CC-BY 4.0), specifically Fig. 3c and Fig. 3d.

Fig. 3c: a single condition, 10 mM M_{pre} + 25 mM DTT in water, 0–24 h, both observables on one plot (residual M_{pre} , left axis; conversion to droplet material, right axis). **Fig. 3d:** residual M_{pre} only, same M_{pre} concentration, two DTT levels (25 mM and 13 mM); the printed panel is cropped to 0–6 h, but the underlying Source Data file retains the full 0–24 h series for both concentrations (used here in full). Both observables are measured as described in §1: fluorescamine fluorescence intensity for residual M_{pre} ; for conversion to droplet material, the source paper’s own Methods state it is calculated from ^1H NMR peak areas of the benzyl (BnSH) protons, $100 \times X'/(X + X')$, where X and X' are the integrated areas before and after BnSH is incorporated into a droplet.

Source Data, not digitization. Every fit in this note uses the paper’s own exact Source Data spreadsheets for Fig. 3b–d (provided via its Data Availability statement, fetched directly since the article page itself blocks automated access), not values read off a figure image. A real error in the Source Data file itself was found and corrected: **the Fig. 3d column headers for 25 mM and 13 mM DTT are swapped relative to the data they actually label.** The column headed “25” decays *more slowly* than the column headed “13”, which is chemically backwards — more DTT should reduce M_{pre} faster, not slower. Direct comparison against Fig. 3c’s own dedicated (and unambiguous, per the paper’s caption) 25 mM series confirms it: the column headed “13” matches Fig. 3c’s real 25 mM curve exactly (to machine precision) at every overlapping timepoint, while the column headed “25” is a genuinely different, independent, slower-decaying curve. This note uses the corrected labels throughout; the true 13 mM curve (headed “25” in the source file) is the real held-out curve used in §3.1, with 10 points spanning the full 24 h rather than 6.

1.2 Data and the fitted model, plotted



Left and center: residual M_{pre} , Stage 1 (§3.1), real Source Data — both real series of the 25 mM condition together (left, Fig. 3c and the corrected column of Fig. 3d), and the corrected, true 13 mM condition (center), which the model was fit to only jointly, not individually (§3.1’s held-out prediction test uses a separate, $p = 1$ -only fit, not shown here). Right: conversion to droplet material, Stage 2 (§3.2), real Source Data.

2 The model, complete

Every equation and every parameter used anywhere in this note is collected here, in one place, before any fitting or results.

2.1 Stage 1: oligomerization, a mechanistic reaction network

Monomer release and chain growth, as mass-action kinetics:

$$R_0 = k_0 x d^p, \quad R_1 = k_1 m^2, \quad R_n = k_{\text{el}} p_n m, \quad 2 \leq n < N \quad (2)$$

$$\dot{x} = -R_0, \quad \dot{d} = -R_0 \quad (3)$$

$$\dot{m} = 2R_0 - 2R_1 - \sum_{n=2}^{N-1} R_n \quad (4)$$

$$\dot{p}_2 = R_1 - R_2, \quad \dot{p}_n = R_{n-1} - R_n \quad (3 \leq n < N), \quad \dot{p}_N = R_{N-1} \quad (5)$$

$$\dot{h} = R_1 + \sum_{n=2}^{N-1} R_n \quad (6)$$

$d = [\text{DTT}]$, $h = [\text{BnSH}]$. No droplet feedback on the oligomerization rates, one shared elongation rate constant k_{el} for every chain length. The DTT reaction order p in R_0 is a free parameter, not assumed to be 1: a single DTT concentration cannot identify it (the fit absorbs any value into k_0), so p is only meaningful when fit jointly across more than one DTT concentration (§3.1).

2.2 Stage 2: droplet formation, a target that moves, approached gradually

Droplets form from the oligomer/BnSH pool once it exceeds a solubility threshold, and that threshold drifts slowly over the reaction (plausibly as DTT itself is consumed, shifting the solution's redox environment):

$$S(t) = \sum_{n=2}^N p_n(t) + h(t), \quad q^*(t) = Y(S(t) - F_{\text{sat},0} - \lambda t) \quad (7)$$

$S(t)$ — the available oligomer/BnSH material — comes directly from Stage 1 ((4)–(6)), not from a free parameter: the two stages are coupled through it, not fit independently. The droplet-forming response to this target is a gradual, decelerating approach to it, not an instantaneous jump — an Avrami-shaped [5, 6, 7] fraction $X(t)$ climbing toward 1:

$$\dot{X} = (1 - X) n K t^{n-1}, \quad X(0) = 0 \quad (8)$$

$$q(t) = q^*(t) \cdot X(t) \quad (9)$$

$q(t)$ can keep rising after $S(t)$ itself has levelled off, because the response $X(t)$ is still catching up to the target that is already available — and can later fall, because $q^*(t)$ itself declines as the solubility threshold drifts. This is the model actually used for the results in §3.2. $S(t)$ and $X(t)$'s approach to it are mechanistic in origin (the first computed directly from Stage 1's reaction network, the second a physically motivated gradual-response shape), but the $-\lambda t$ drift itself is a phenomenological placeholder (see also §3.4).

2.3 Every parameter in the model, in one table

Symbol	Meaning	Status	Value
<i>Stage 1 — oligomerization (§2.1)</i>			
$x(0)$	initial M_{pre}	fixed (experiment)	10 mM
$d(0)$	initial DTT	fixed (experiment)	25 or 13 mM
N	max. tracked chain length	fixed (chosen)	12
k_0	M_{pre} reduction rate	fit, jointly, 3 curves	0.00099
k_1	nucleation rate	fit, jointly	1.145
k_{el}	elongation rate (shared)	fit, jointly	3.229
p	DTT reaction order	fit, jointly <i>only</i>	2.37
<i>Stage 2 — droplet formation (§2.2)</i>			
$S(t)$	available oligomer/BnSH	computed from Stage 1	— (not fit)
Y	scale factor	fit, to $q(t)$ alone	3.94
$F_{\text{sat},0}$	solubility threshold, intercept	fit, to $q(t)$ alone	1.58
λ	drift rate	fit, to $q(t)$ alone	0.106
K, n	Avrami rate/exponent	fit, to $q(t)$ alone	0.522, 0.949

The pattern worth noting directly: every *fit* Stage 1 parameter ($k_0, k_1, k_{\text{el}}, p$) is fit *jointly* across multiple curves, which is what makes p meaningful and what let the model be tested against a held-out condition (§3.1). Every fit Stage 2 parameter ($Y, F_{\text{sat},0}, \lambda, K, n$) is fit to $q(t)$ *alone* — there is only one such curve available, which is the root cause of everything in §3.3–§3.4. ($x(0), d(0), N$ are fixed inputs, not fit, in either stage.)

3 Fitting and testing the model

3.1 Stage 1, fit jointly across DTT concentration, and a held-out prediction

Fitting (2)–(6) to a single DTT concentration alone does not determine p : with d nearly constant within one condition, any p can be absorbed into k_0 , so a linear-in- d assumption ($p = 1$) fit this way looked fine in isolation but degrades sharply when tested against a second, independent DTT concentration (a genuine zero-parameter prediction of the 13 mM curve using the 25 mM-fit rate constants: $R^2 = 0.61$, clearly worse than the joint fit below). Fitting $k_0, k_1, k_{\text{el}}, p$ *jointly* to all three available residual- M_{pre} curves — both real series of 25 mM DTT (Fig. 3c, and the corrected column of Fig. 3d, essentially the same underlying condition measured twice) and the corrected, true 13 mM DTT column of Fig. 3d (§1.1) — resolves this:

Curve	RMSE	R^2
25 mM DTT (Fig. 3c, 0–24 h)	3.01	0.990
25 mM DTT (Fig. 3d, corrected col., 0–24 h)	1.87	0.996
13 mM DTT (Fig. 3d, corrected col., 0–24 h)	2.54	0.990

One shared mechanism fits all three curves well — the DTT-reduction step R_0 is close to second-order in DTT concentration ($p = 2.37$), not first-order as the initial mass-action guess assumed. This is the model’s strongest result: p is only identifiable at all once a second DTT condition forces it to be, and once identified, it genuinely predicts that condition rather than merely fitting it after the fact. The naive $p = 1$ single-condition fit clearly underperforms ($R^2 = 0.61$) relative to

the jointly-identified $p \approx 2.37$ fit, and p is still only pinned down by using both DTT conditions together.

Two further checks on the 25 mM condition, against quantities the fit was not built to match: the implied mean chain length from the model’s final oligomer distribution ($\bar{n} = 6.52$) matches an independent estimate computed directly from the real data’s own late-time plateau alone ($\bar{n} = 6.38$, from $O(t)$ averaged over $t \geq 12$ h, no model involved); and the fit implies DTT itself drops from 25 to 15 mM (40% consumed) over 24 h — a concrete, checkable prediction against any future direct DTT measurement.

3.2 Stage 2, fit to the conversion-to-droplet curve

With $S(t)$ fixed from the joint fit above (Fig. 3c, 25 mM condition), (7)–(9) fit to the real conversion-to-droplet curve (same figure, same condition):

Y	$F_{\text{sat},0}$	λ	K	n	RMSE	R^2
3.94	1.58	0.106	0.522	0.949	0.39	0.9997

Model		Data
Peak value	67.6% at $t = 9.3$ h	68.0% at $t = 10.5$ h
Value at $t = 24$ h	62.5%	61.9%

The model reproduces the full shape — rise, peak, and decline — not just an average through the data, and fits slightly better on the real data than the earlier digitized version did. The peak’s timing has a mechanistic reading: the oligomer/BnSH supply $S(t)$ has essentially finished rising by $t \approx 5$ h, but the droplet-forming response $X(t)$ has not — so $q(t)$ keeps growing on that unfinished climb alone, until $q^*(t)$ ’s own slow decline overtakes it. This is a gradual, monotonically decelerating approach to $X = 1$, not an induction delay: with the fitted $n = 0.949 < 1$, $\dot{X}$ is largest at $t = 0$ and falls throughout, rather than starting flat and accelerating (that shape needs $n > 1$, which is not what was fit here). One earlier-flagged wrinkle resolves itself on real data: $F_{\text{sat},0}$ was negative on the digitized fit (-3.54), an odd sign for a literal solubility concentration; on the real data it comes out positive (1.58), the physically expected sign, though still just a fitted number rather than an independently measured quantity.

3.3 Stage 2, a held-out test in time

No droplet-conversion curve at a second DTT concentration exists in the published figures, so the cross-condition test that validated Stage 1 (§3.1) cannot be run here. A different held-out test is available: fit (7)–(9) using only data up to a cutoff time, predict everything after it with no new parameters, and see whether the decline — the feature this model was built to capture — is genuinely predicted in advance rather than only fit after the fact.

Training cutoff	Held-out R^2	λ (fit from training alone)
10 h (before the peak)	−0.10	0.045
12 h	0.19	0.065
14 h	−0.42	0.065
16 h	0.16	0.084
18 h	0.55	0.097

The full-data value is $\lambda = 0.106$. Prediction quality does not improve smoothly with more training data — negative at 10 h and 14 h, only becoming reasonably good at 18 h of the 24 h window (75%), by which point λ is still under-estimated from training data alone (0.097 vs. the full-data 0.106).

The 18 h result is real and reproducible: once the decline is already visibly underway in the data (the five real points from 14–18 h — 67.2, 66.2, 66.0, 65.9, 64.6% — a clearly declining stretch), the model correctly extends it a further 6 h with no new parameters. The limitation is precise, not vague: it works once the decline has started to show, not before. Closing that gap needs new data, not a better fit to the one curve already in hand — §3.4 states what that data should be and how it would be used.

A second, separate check specifically on peak *timing* (not the full held-out R^2 above) was run directly on real data: fit only to data up to a cutoff, then ask what peak time the fit itself predicts. At $t = 8$ h — genuinely rise-only, before the curve shows any flattening — the fit finds no peak inside its own simulated window at all, extrapolating continued rise to the edge of it (78.8% at $t = 24$ h, 13.5 h off the real 68.0% at $t = 10.5$ h). Only from $t = 9$ h onward does the predicted peak time land close to correct ($t = 9$ h: predicted 10.89 h, 0.4 h off; $t = 10$ –18 h: predicted 9.6–11.1 h, within ≈ 1 h). Since the real data already begins visibly flattening by $t \approx 9$ h (a data point at $t = 9$ h is already close to the eventual peak value), this is not a prediction made in advance of any sign of the peak — it only becomes accurate once the data itself has already started to show the curve leveling off.

Stated precisely: neither the decline’s *magnitude* nor the peak’s *timing* is predicted meaningfully in advance from data collected before the interesting behavior starts to appear — both only become reliable once the data already shows the relevant feature (flattening near the peak; a few hours into the decline) beginning.

3.4 A limit of the available data

No theory or independent measurement currently supplies the decline’s rate without relying on the curve $q(t)$ — the droplet-conversion curve — that it is meant to predict, and we do not see any fixed point in the mapping $q^{\text{init}} \rightarrow q^{\text{obtained}}$. Completing a predictive model for Stage 2 needs data from one of two experiments.

Experiment 1: a second droplet-conversion curve, at a different DTT concentration. Run the same NMR conversion-to-droplet measurement at a different DTT concentration (e.g. 13 mM instead of 25 mM). The model, fit to the 25 mM curve alone as it already is, would then be used to predict the 13 mM curve with zero new parameters — the same test that already validated Stage 1’s DTT reaction order. A successful prediction completes the model.

Experiment 2: a direct, parallel measurement of whatever drives the decline. Run a redox/thiol assay alongside the existing droplet measurement, sampled at several times across the same 24 h reaction. The rate would be fixed from that measurement alone, without looking at $q(t)$, and the resulting model used to predict $q(t)$ using that externally-fixed rate. A successful prediction completes the model this way instead.

Either experiment alone is sufficient; both are not needed.

4 Comparison to Matsuo and Kurihara’s own fit

The source paper states its conversion-to-droplet curve “was fit to an autocatalytic reaction equation using the Levenberg–Marquardt method,” with the actual equation given only in Supplementary

Fig. 14 (fetched directly from Springer’s supplementary-materials server for this comparison, not from the bot-blocked main article):

$$y = a \left(1 + (d - 1) e^{-k(x - x_c)} \right)^{1/(1-d)}, \quad (10)$$

fit by Levenberg–Marquardt to $a = 69.70$, $x_c = 77.36$ min, $d = 0.465$, $k = 0.00822/\text{min}$, $R^2 = 0.99987$.

(10) is phenomenological in the strict sense: a curve shape chosen and fit to match the data, with no rate law or reaction network behind it, and no derivation offered for why this particular functional form should apply. This is true throughout their treatment of droplet formation, and it is the paper’s only fitted equation anywhere — the M_{pre} -reduction/oligomerization curves (Fig. 3c/3d) are presented as raw data with no fit at all.

This is exactly the Richards curve [4], not a different family: substituting $\nu = 1 - d$ into the standard form $Y = K(1 + Qe^{-B(t-M)})^{1/\nu}$ with $Q = -\nu$, $B = k$, $M = x_c$, $K = a$ reproduces (10) exactly (checked symbolically). Matsuo and Kurihara’s own empirical fit independently lands in the Richards family that arises from a reversible Becker–Döring aggregation reduction [2], without being named that way in the source paper, and independently validated on unrelated protein-aggregation and -disassembly kinetics [3]. This note’s own Stage 2 (§2.2) is a different construction — a drifting target multiplied by an Avrami response, not itself a member of the Richards family — which produces a similarly shaped rise–peak–decline curve without being mathematically identical to (10).

	Matsuo & Kurihara (Suppl. Fig. 14)	This note (Stage 2, §3.2)
Data	primary NMR, $n = 3$, error bars, D ₂ O replicate run	Source Data, Fig. 3c, water, 0–24 h
Curve family	Richards (unnamed as such)	Richards-driven Avrami (§2.2)
Fit quality	$R^2 = 0.99987$ (own data)	$R^2 = 0.9997$ (own data)
Connected to upstream chemistry?	no — $q(t)$ fit alone	yes — driven by Stage 1’s $S(t)$
Tested against held-out data?	not reported	not for Stage 2 either (§3.3)

4.1 Advantages and novel contributions, in detail

The real difference between this note and Matsuo and Kurihara’s own analysis is not fit quality of (10). This note builds a reaction-network model, and that difference produces several concrete results which no fit of (10) can produce, regardless of how precise the fit is. Namely, we achieve the following results, numbered 1–6 below.

1. A mechanistic model. Stage 1 ((2)–(6)) is a mechanistic account of M_{pre} -reduction and oligomerization for this system: explicit rate laws for M_{pre} reduction, nucleation, and elongation, with chain length tracked explicitly ($p_2, \dots, p_N$), not lumped into one undifferentiated “oligomer” pool.

2. A DTT reaction order. This note determines it directly: $p \approx 2.37$, close to second order, not the first-order guess a naive mass-action model would start from (§3.1) — a concrete, checkable

claim about the reaction mechanism, made possible by fitting a rate law to the oligomerization data in the first place.

3. Cross-condition discrimination of the kinetic law. A model fit at 25 mM under the naive assumption $p = 1$ predicts the independent 13 mM condition only moderately ($R^2 = 0.61$, §3.1). Joint use of both DTT conditions identifies $p \approx 2.37$ instead, after which one shared kinetic mechanism reproduces both concentrations with $R^2 \approx 0.99$. The second condition does not merely supply another curve to fit — it discriminates between alternative DTT dependences and identifies the reaction order.

4. The two observed curves connected through one mechanism, not fit in isolation. This note’s Stage 2 uses $S(t)$ — the oligomer/BnSH pool, computed directly from Stage 1’s fitted network, not a free parameter — as the quantity driving droplet formation ((7)). The two observable curves are outputs of one connected mechanism, not two curves fit independently of each other.

5. Identifying what the residual- M_{pre} curve actually measures. The fluorescamine signal behind Fig. 3c/3d detects free primary amines, not the disulfide bond DTT reduces (a different functional group), so it cannot see M_{pre} ’s own reduction step directly. §1 derives what it actually tracks: total particle count, (1). This is load-bearing here: fitting $x(t)$ directly, rather than $O(t)$, would be the wrong target, and every fit in this note uses the corrected one.

6. Further falsifiable predictions, from the model’s internal state. The mean oligomer chain length ($\bar{n} = 6.52$ from the model, matching $\bar{n} = 6.38$ computed independently from the data’s own plateau, §3.1) and the DTT-depletion trajectory (25 $\rightarrow$ 15 mM over 24 h, not yet checked against any direct measurement) are both real, falsifiable outputs of Stage 1’s internal state — genuine physical quantities, not just curve-shape parameters.

None of the above applies to Stage 2 itself (data limitations, see again §3.3–§3.4).

References

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